VACATION MINI PROJECT REPORT

PENNYPILOT

A Feature-Rich Client-Side Personal Finance and Expense Tracking Application

Project Report submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Computer Applications (BCA)

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Submission	July 2026

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CERTIFICATE

This is to certify that the mini project entitled

PENNYPILOT — EXPENSE TRACKER

has been successfully developed and submitted by **SANJAY** (Roll No. **24BCA082**), a student of the 4th Semester, Bachelor of Computer Applications (BCA), St. Claret College (Autonomous), Bengaluru, in partial fulfillment of the academic requirements for the degree of Bachelor of Computer Applications.

This mini project was developed independently as part of self-learning and skill enhancement during the vacation period. The work presented in this report is original and has not been submitted elsewhere for any academic examination or award.

The project demonstrates a commendable level of initiative and competence in front-end web development, encompassing financial data management, interactive data visualisation, and modern UI design.

St. Claret College (Autonomous), Bengaluru

DECLARATION

I, **SANJAY**, student of 4th Semester, Bachelor of Computer Applications (BCA), St. Claret College (Autonomous), Bengaluru (Roll No. 24BCA082), hereby declare that the mini project report entitled **"PENNYPILOT — EXPENSE TRACKER"** submitted to the Department of BCA is a record of original work done by me during the vacation period.

I further declare that:

- This project has been carried out independently, without any formal institutional guidance, as a self-initiated learning exercise.
- The work presented in this report is genuine and has not been copied or reproduced from any other source without proper acknowledgement.
- This project has not been submitted previously for any degree, diploma, or academic title at this or any other institution.
- All references and sources of information used have been duly cited and acknowledged.
- The analysis performed and the conclusions drawn are based entirely on my own understanding and research.

Place: Bengaluru

Date: July 2026

Signature of the Student

SANJAY

Roll No: 24BCA082

4th Semester, BCA

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who encouraged and supported me throughout the development of this mini project.

I am grateful to the vast community of developers and educators whose open resources — including MDN Web Docs, W3Schools, Stack Overflow, and YouTube tutorials — provided essential guidance during development. These platforms helped me understand complex JavaScript patterns, Chart.js integration, and responsive CSS layout techniques.

I sincerely thank my family for their unwavering support, patience, and encouragement throughout the project. Their motivation provided the drive needed to complete this ambitious application during the vacation period.

This project was undertaken entirely on my own initiative. Building a full-featured personal finance application from scratch — complete with multi-module navigation, interactive charts, voice input, and data export — has been one of the most enriching technical experiences of my academic journey so far.

SANJAY

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ABSTRACT

Managing personal finances is a fundamental life skill, yet one that a large proportion of young adults — including students — find challenging in practice. The absence of a dedicated, easy-to-use tool often leads to untracked expenditure, poor budgeting habits, and financial stress. Existing personal finance applications are either overly complex, subscription-based, or require cloud account creation, creating barriers for students seeking a simple, private solution.

PennyPilot is a comprehensive, client-side personal finance and expense tracking web application designed to help users take control of their financial lives. Built using HTML5, CSS3, and JavaScript — with Chart.js for data visualisation — it operates entirely in the browser without any server-side infrastructure, login, or subscription.

Key Features:

- Dashboard with real-time financial summaries: total balance, income, expenses, savings rate, and monthly spending.
- Transaction management: add income and expense entries with category, payment method, recurrence, receipt attachment, and notes.
- Budget module: set and track monthly spending limits per category with visual progress indicators.
- Analytics module: expense breakdown pie chart, 6-month income vs expense trend, and category comparison bar chart.
- Insights module: AI-style personalised financial tips, savings rate evaluation, and spending pattern alerts.
- Calendar view for date-based transaction browsing.
- Voice Input: add transactions using natural language speech recognition.
- Goals & Achievements: savings streak tracking and gamified achievement badges.
- Export/Import: download transactions as JSON, CSV, or PDF; restore from JSON backup.
- Light/Dark mode toggle for personalised visual comfort.

Technologies Used:

HTML5, CSS3, Vanilla JavaScript, Chart.js, Web Speech API, localStorage API. Deployed on Netlify.

Expected Outcome:

PennyPilot is expected to help users develop disciplined financial habits, gain meaningful insights into their spending patterns, and achieve their savings goals — all within a privacy-first, zero-setup platform accessible from any device.

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CHAPTER 1

INTRODUCTION

1.1 Background

Financial literacy and personal budgeting are increasingly recognised as essential life skills, particularly for students transitioning into independent living. The ability to track income and expenses, set budgets, and understand spending patterns is directly linked to financial wellbeing. Despite this, many individuals lack the tools or discipline to maintain consistent financial records.

Digital personal finance tools have proliferated in recent years, yet most cater to advanced users or require cloud accounts, bank linking, and recurring fees. A lightweight, private, browser-based alternative that covers the full spectrum of personal finance management without these barriers represents a significant opportunity.

1.2 About the Project

PennyPilot (tagline: *expense tracker*) is a fully client-side personal finance web application offering nine functional modules within a single, beautifully designed interface. It supports transaction logging, category budgeting, multi-chart analytics, AI-style financial insights, calendar browsing, voice input, gamified goals, and data export — all without any server communication or user account.

The application is live at: **https://penny-pilot-track.netlify.app/**, deployed via Netlify. It supports both dark and light modes and is fully responsive across devices.

1.3 Scope

- Complete income and expense transaction lifecycle management.
- Category-based budget setting and real-time expenditure tracking.
- Multi-dimensional financial analytics using interactive charts.
- AI-style personalised insights based on spending patterns.
- Voice-controlled transaction entry via the Web Speech API.
- Gamified savings goals and achievement badges.
- Data portability through JSON, CSV, and PDF export.

CHAPTER 2

OBJECTIVES OF THE PROJECT

The following objectives guided the design and development of PennyPilot:

1. To provide comprehensive personal finance tracking: Enable users to log all income and expense transactions with rich metadata — category, payment method, recurrence, notes, and receipt images — giving a complete picture of their finances.

2. To implement category-based budgeting: Allow users to set monthly spending limits per category and receive real-time visual feedback on budget utilisation through colour-coded progress bars.

3. To deliver meaningful financial analytics: Render multi-dimensional charts — expense breakdown, 6-month trend, income vs expenses, and category comparison — helping users understand their financial patterns at a glance.

4. To generate personalised financial insights: Analyse transaction data to produce contextual tips and alerts, such as savings rate evaluation and top spending category identification, comparable to AI-driven financial advisors.

5. To incorporate innovative input methods: Implement voice-controlled transaction entry using the Web Speech API, enabling hands-free expense logging through natural language commands.

6. To gamify financial discipline: Introduce a daily logging streak counter and achievement badge system to reward consistent financial tracking behaviour and sustain user engagement.

7. To ensure data portability and privacy: Provide export functionality in JSON, CSV, and PDF formats and import via JSON backup, while keeping all data local to the user's browser with no server transmission.

CHAPTER 3

EXISTING SYSTEM

3.1 Current Approaches to Personal Finance Management

Users currently rely on a range of tools and methods to manage personal finances, each carrying significant limitations when evaluated against a student's practical needs:

a) Manual Record-Keeping

Notebooks, diaries, and handwritten ledgers remain popular among students for their simplicity. However, they offer no computation, no visualisation, no search, and no data persistence beyond the physical document. Errors are common and analysis is impractical.

b) Generic Spreadsheets

Google Sheets and Microsoft Excel provide computation and charting capabilities but require manual formula construction and offer no financial-domain-specific features such as category budgets, insights, or voice input. The learning curve is steep for non-technical users.

c) Cloud-Based Finance Apps

Applications such as Mint, YNAB, or Money Manager require account registration, internet connectivity, and — in most cases — bank account linking or paid subscriptions. Privacy-conscious users are reluctant to share sensitive financial data with third-party servers.

d) Mobile Expense Apps

Many mobile apps offer basic expense tracking but are platform-specific (Android/iOS), lack web access, and often include intrusive advertisements or in-app purchases that limit functionality.

3.2 Limitations of Existing Solutions

- Require cloud accounts and continuous internet connectivity.
- Involve sharing sensitive financial data with third-party servers.
- Lack integration of analytics, insights, voice input, and goals in a single tool.
- Subscription or payment models create access barriers for students.
- No gamification mechanisms to encourage consistent financial tracking.
- Limited export flexibility; data portability is often restricted.
- Platform dependency: mobile-only or desktop-only applications.

CHAPTER 4

PROPOSED SYSTEM

4.1 Overview

PennyPilot proposes a fully integrated, client-side personal finance application that consolidates transaction management, budget tracking, visual analytics, AI-style insights, voice entry, and gamified goal-setting into a single, privacy-first platform. It eliminates every friction point of existing solutions — no account, no server, no subscription, no installation — while delivering a feature set that rivals commercial applications.

4.2 Advantages Over the Existing System

- **Complete Privacy:** All financial data is stored exclusively in the user's browser via localStorage. No data is transmitted to any server or third party.
- **Zero Barriers to Entry:** Open the URL and begin tracking immediately. No registration, login, or installation required.
- **All-in-One Platform:** Nine functional modules — Dashboard, Transactions, Budgets, Analytics, Insights, Calendar, Voice Input, Goals, Export/Import — available within a single application.
- **Voice-Controlled Entry:** Natural language speech recognition for hands-free transaction logging is a capability absent from most student-targeted tools.
- **Rich Data Visualisation:** Multiple chart types (pie, line, bar, doughnut) provide multi-dimensional financial analysis without requiring external software.
- **Gamification:** Daily streaks and achievement badges create a reward loop that promotes consistent financial tracking habits.
- **Data Portability:** Full export to JSON, CSV, and PDF gives users complete control over their data and enables backup and migration.

4.3 System Architecture

PennyPilot is a single-page application (SPA) with a persistent sidebar navigation and a dynamic main content area that renders one module at a time. All application state, transaction data, budget configurations, and goal progress are persisted via the browser's localStorage API. Chart.js is used for all data visualisations. The Web Speech API powers voice input. The application is distributed as a set of static files hosted on Netlify's global CDN.

CHAPTER 5

MODULES AND FEATURES

PennyPilot comprises nine functional modules accessible through a persistent left-hand sidebar. Each module is described in detail below.

5.1 Dashboard Module

The Dashboard is the application's home view and provides a comprehensive at-a-glance financial summary for the user.

Summary Cards:

Four prominent cards display Total Balance, Total Income, Total Expenses (all time), and Savings with the savings rate percentage. A fifth card shows the current month's total spending.

Charts:

- Spending by Category: a doughnut chart breaking down expenditure across Food, Transport, Shopping, Bills, Entertainment, Health, and Education.
- Income vs Expenses: a grouped bar chart comparing monthly income and expense totals over the past six months.
- Monthly Trend: a line chart tracking expense progression month-over-month.
- Weekly Spending: a bar chart showing daily spending for the current week.

Recent Transactions:

A quick-access table at the bottom of the dashboard lists the most recent transactions with a 'View all' link to the full Transactions module.

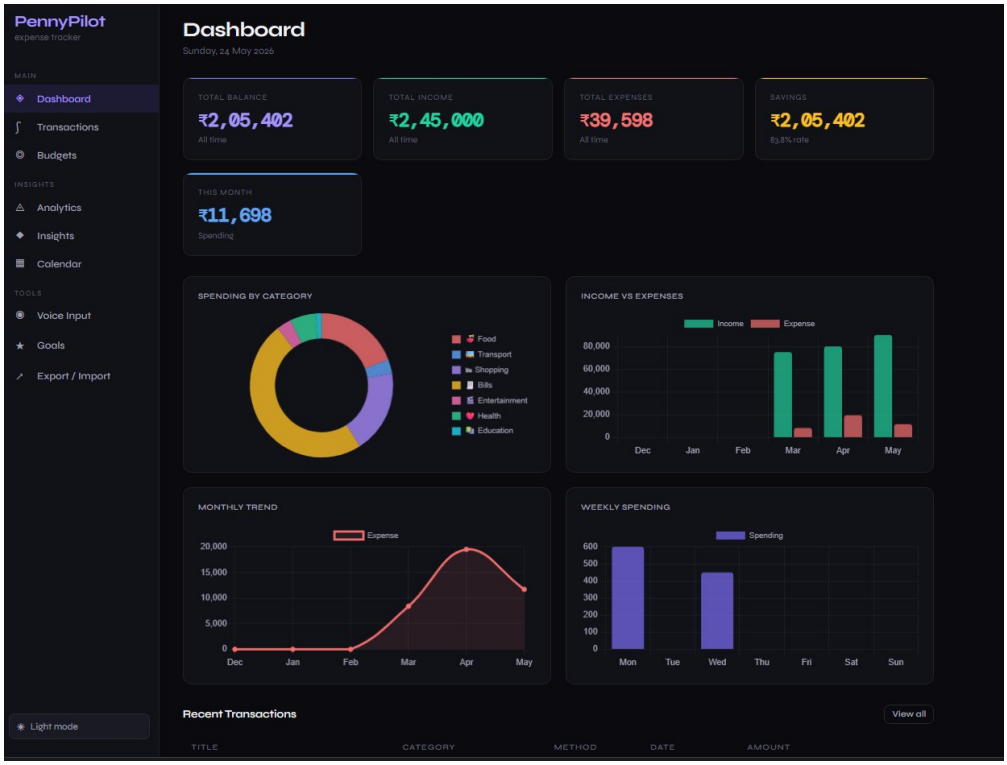


Figure 5.1 – Dashboard Module with Financial Summary Cards and Charts

5.2 Transactions Module

The Transactions module is the data entry core of PennyPilot. It provides a full-featured interface for logging, browsing, and managing all financial records.

Add Transaction Form:

A modal form allows entry of income or expense transactions with the following fields: title, amount (■), category, date, payment method (Cash/UPI/Card/Net Banking/Wallet), recurrence (None/Daily/Weekly/Monthly/Yearly), notes, and an optional receipt image attachment.

Filtering and Sorting:

Transactions can be filtered by type (income/expense), category, and payment method. Sorting options include latest first, oldest first, highest amount, and lowest amount. A search bar enables quick lookup by title or keyword.

5.3 Budgets Module

The Budgets module enables users to define monthly spending limits for each expenditure category and monitor progress in real time.

Each category card displays the amount spent against the set limit (e.g., ■3,250 / ■8,000 for Food) with a colour-coded horizontal progress bar. Categories without a set limit show a 'Set limit' button. A global '+ Set budget' button opens the budget configuration modal. Categories covered include Food, Transport, Shopping, Bills, Entertainment, Health, Education, and Other.

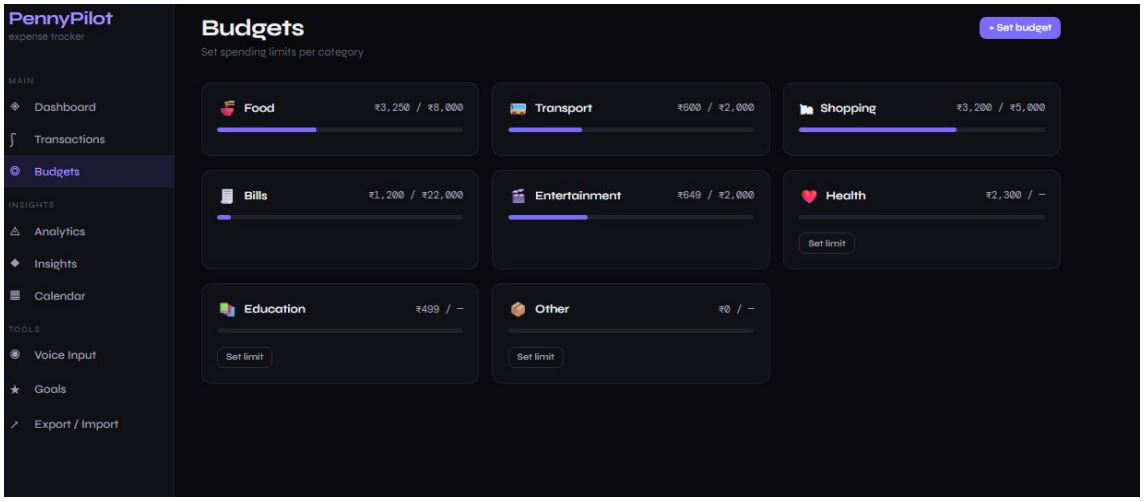


Figure 5.2 – Budgets Module with Per-Category Spending Limits and Progress Bars

5.4 Analytics Module

The Analytics module provides a deep-dive view of financial data through four distinct chart visualisations:

- **Expense Breakdown:** A pie chart illustrating the proportional share of each spending category in total expenditure.
- **6-Month Trend:** A dual-line chart overlaying income and expense trajectories across the past six months, revealing patterns of growth or overspending.
- **Income vs Expenses (6M):** A grouped bar chart comparing monthly income and expense totals side-by-side for the six-month period.
- **Category Comparison:** A horizontal bar chart ranking all categories by total spend, enabling quick identification of the highest-cost areas.



Figure 5.3 – Analytics Module with Expense Breakdown, Trend, and Category Comparison Charts

5.5 Insights Module

The Insights module performs smart analysis of the user's financial data and presents key performance indicators alongside contextual, personalised recommendations.

KPI Cards:

Four metric cards display: Savings Rate (% of total income saved), Average Daily Spend, Top Spending Category (with total), and Total Transactions (split by income and expense count).

Smart Insights:

Below the KPIs, the module displays contextual insight messages generated from the user's data — for example: confirming that food spending is within a healthy range, congratulating a savings rate exceeding the recommended 20%, and summarising the current month's total spending. These messages adapt dynamically as transaction data changes.

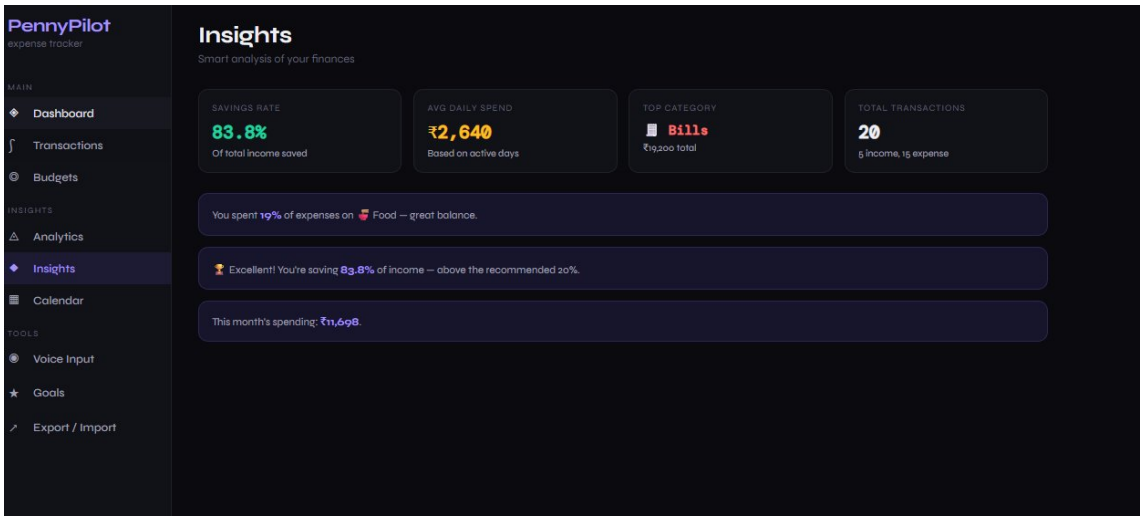


Figure 5.4 – Insights Module with KPI Cards and Personalised Financial Tips

5.6 Calendar Module

The Calendar module presents a month-view calendar with navigation arrows for browsing previous and future months. Dates on which transactions were recorded are highlighted, allowing users to select any date and view the transactions that occurred on that day. This provides an intuitive, date-centric perspective on financial activity.

5.7 Voice Input Module

The Voice Input module leverages the browser's Web Speech API to enable hands-free transaction entry. A microphone button activates speech recognition; the user speaks a natural language command such as 'Spent 250 rupees on food' or 'Received 50,000 salary', and the application parses the utterance to extract the amount, type, and category before creating the transaction automatically. Example prompts are displayed on-screen to guide first-time users.

5.8 Goals and Achievements Module

The Goals module gamifies the process of financial tracking by maintaining a daily logging streak counter — rewarding users who log transactions every consecutive day. An Achievements section displays a collection of milestone badges (e.g., first transaction, 7-day streak, 100 transactions) that unlock as users reach specific milestones, creating a reward loop that encourages habitual engagement with the application.

5.9 Export and Import Module

The Export/Import module ensures full data portability. Users can download their transaction data in three formats: a JSON backup (complete data including budgets), a CSV export (transactions as a spreadsheet for use in Excel or Sheets), and a PDF Summary (a formatted monthly report suitable for printing). Data can be restored from a previously downloaded JSON backup by dragging and dropping the file or using the browse dialog.

CHAPTER 6

TECHNOLOGIES USED

PennyPilot was developed using the following front-end technologies. No back-end server, database, or external API is used for core functionality.

6.1 HTML5

HTML5 provides the structural framework for PennyPilot's nine-module layout. Semantic elements define the sidebar, main content area, modal dialogs, and form controls. HTML5 APIs — including the File API for receipt attachment and the drag-and-drop interface for JSON import — are used to enhance user interactions without additional libraries.

6.2 CSS3

CSS3 delivers PennyPilot's polished dark-themed interface, characterised by a deep navy-black background with indigo/purple accents and vibrant chart colours. CSS Custom Properties power both the dark and light mode themes through a single variable swap. Flexbox and CSS Grid manage the responsive multi-column layouts, while CSS transitions animate modal appearances and sidebar highlights.

6.3 JavaScript (ES6+, Vanilla)

All application logic is implemented in Vanilla JavaScript. Key implementations include: modular function-based architecture for each module, transaction CRUD operations with localStorage synchronisation, dynamic chart rendering driven by real-time data, speech recognition event handling, CSV and JSON serialisation for export, and PDF generation for the monthly summary report.

6.4 Chart.js

Chart.js is the sole external library used in the project. It renders the application's interactive charts — doughnut, pie, line, and bar types — with responsive sizing, animated transitions, and customisable colour palettes that match the application's theme. Chart instances are updated dynamically whenever transaction data changes.

6.5 Web Speech API

The browser-native Web Speech API's SpeechRecognition interface powers the Voice Input module. It converts spoken language to text in real time, which is then parsed using JavaScript pattern matching to extract transaction details from natural language utterances.

6.6 localStorage API

The Web Storage API's localStorage interface provides persistent key-value storage within the browser. Transactions, budgets, goals, and application settings are serialised as JSON and stored locally, surviving page refreshes and browser restarts without any server interaction.

6.7 Netlify

The application is deployed on Netlify's static hosting platform, providing global CDN distribution, automatic HTTPS, and a custom subdomain. The deployment requires only the upload of static HTML, CSS, and JavaScript files.

#	Technology	Type	Purpose
1	HTML5	Markup	Structure, forms, file/drag-drop APIs
2	CSS3	Styling	Dark/light theme, layout, animations
3	JavaScript ES6+	Programming	App logic, DOM, localStorage, export
4	Chart.js	Library	Interactive financial charts
5	Web Speech API	Browser API	Voice-controlled transaction entry
6	localStorage	Browser API	Client-side data persistence
7	Netlify	Platform	Static hosting and CDN deployment

CHAPTER 7

CONCLUSION

PennyPilot represents a highly ambitious and successful mini project that goes well beyond the typical scope of a student vacation exercise. By delivering nine fully functional modules — spanning transaction management, budget tracking, multi-chart analytics, personalised insights, voice input, gamified goals, and data export — within a single, privacy-first, client-side web application, the project demonstrates exceptional initiative and technical depth.

From a technical perspective, the project showcases the power of the modern browser platform. The integration of Chart.js for rich data visualisation, the Web Speech API for voice-controlled input, and the localStorage API for persistent offline storage — all orchestrated through Vanilla JavaScript without any back-end infrastructure — illustrates what can be achieved through a thorough understanding of the core web platform.

The inclusion of a gamification layer through daily streaks and achievement badges reflects an understanding of user psychology and engagement design that extends beyond pure technical implementation. Similarly, the transparent handling of data — with explicit export options and local-only storage — reflects a principled, user-first approach to software design.

On a personal level, independently designing, developing, and deploying an application of this complexity during the vacation period has been a transformative learning experience. It has deepened practical skills in JavaScript architecture, responsive UI design, data visualisation, and browser API integration — competencies that will be directly applicable throughout the remainder of the BCA programme and in professional software development.

In conclusion, PennyPilot is a polished, feature-complete, and publicly accessible personal finance application that fully achieves its stated objectives and stands as a strong demonstration of the developer's growing technical competence and product thinking.

Live Project: https://penny-pilot-track.netlify.app/